

TECHNOLOGICAL DESIGN

Technological design courses provide students with a variety of learning experiences that focus on the practical application of the principles of engineering, architecture, and design. These activity-based courses emphasize problem solving to meet design challenges in a wide range of areas, which may include apparel and textile design, architectural design, interior design, mechanical and industrial design, and robotics and control systems.

Students learn to apply knowledge of research, historical trends, design, materials, fabrication methods, and testing criteria to develop innovative and environmentally sustainable products, processes, and/or services. The technologies and processes used to create design solutions may include both traditional and computer-based drafting methods, scale models, working prototypes, animations and simulations, displays, portfolios, and presentations.

The list of approved emphasis areas for technological design can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Technological Design, Grade 11

University/College Preparation

TDJ3M

This course examines how technological design is influenced by human, environmental, financial, and material requirements and resources. Students will research, design, build, and assess solutions that meet specific human needs, using working drawings and other communication methods to present their design ideas. They will develop an awareness of environmental, societal, and cultural issues related to technological design, and will explore career opportunities in the field, as well as the college and/or university program requirements for them.

Prerequisite: None

A. TECHNOLOGICAL DESIGN FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of factors and relationships that affect technological design and the design process;
- A2.** describe appropriate strategies, techniques, and tools for researching, organizing, planning, and managing design projects and related activities, with an emphasis on financial, human, and material resources;
- A3.** demonstrate an understanding of drafting standards, conventions, and guidelines for various types of drawings used to represent designs;
- A4.** demonstrate an understanding of a variety of tools, materials, equipment, and processes used to build, test, and evaluate models and prototypes;
- A5.** use appropriate terminology and communication methods to document, report, and present progress and results.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe ways in which society, the environment, and the economy inspire and/or affect technological design (*e.g., need for barrier-free access or alternative-energy vehicles*), with reference to key technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) (see pp. 7–8);
- A1.2** identify the steps in the design process (*e.g., define the problem or challenge, taking into account relevant contextual or background information; gather information about criteria, constraints, and available materials; generate possible solutions, using techniques such as brainstorming; choose the best solution; develop and produce a model or prototype; test the model or prototype; incorporate improvements or redesign and retest; report results*) (see pp. 22–23), and demonstrate an understanding of the relationships among the steps (*e.g., prototype testing can show that more idea development is needed*);
- A1.3** evaluate various technical products (*e.g., portable music player, backpack, cell phone*) in terms of key technological concepts;

A1.4 demonstrate an understanding of the relationship between technological design and art, science, technology, the environment, and commerce;

A1.5 establish design criteria based on client interviews, consultations, and research (*e.g., determine client's needs, budget, and lifestyle; research technical requirements, material limitations, function of space, and existing infrastructure*);

A1.6 describe the role that the five senses (touch, taste, smell, sight, and hearing) play in technological design (*e.g., shapes, colours, and layouts that make controls easy to read and operate; lighting and colour schemes that set a mood and/or make a room, garden, or building attractive*).

A2. Research and Project Management

By the end of this course, students will:

- A2.1** identify and locate sources of technical and design information (*e.g., Machinery's Handbook, Sweet's building product catalogues, magazines, the Internet, interviews*), using a variety of techniques and tools;
- A2.2** research and describe strategies for the planning, organization, and management of human, material, and financial resources for a design project (*e.g., strategies for delegating tasks, selecting materials, estimating costs*).

A3. Representing Design Ideas Graphically

By the end of this course, students will:

A3.1 identify and describe different methods for representing design ideas graphically (e.g., *mind maps, sketches, design layouts, computer-aided drafting*), with reference to principles and elements of graphic design (e.g., **principles:** *gradation, emphasis, pattern, balance*; **elements:** *space, shape, size, value*);

A3.2 demonstrate an understanding of drawing types (e.g., *pictorial drawings, floor plans, elevations, sections, detail drawings, rendered drawings*) and of drafting standards and conventions (e.g., **standards:** *Canadian Standards Association [CSA]*; **conventions:** *symbols, abbreviations, shading, dimension labels, geometries*), with an emphasis on working drawings;

A3.3 accurately interpret technical drawings and specifications.

A4. Making and Testing Models and Prototypes

By the end of this course, students will:

A4.1 identify and describe different types of models (e.g., *conceptual, physical, virtual, theoretical*) and prototypes (e.g., *proof-of-principle, functional, form study*);

A4.2 compare a variety of modelling tools (e.g., *shaping tools; fabric shears; computer-aided design [CAD] software; computer numerically controlled [CNC] mill, lathe, or router*) and materials (e.g., *matt board, starch, linen*) in terms of suitability, time, budget, and availability;

A4.3 identify various criteria for assessing models and prototypes (e.g., *adherence to specifications, material cost, assembly time, material availability, waste produced, cultural appropriateness*) and the methods and equipment used to perform the assessment (e.g., **methods:** *measuring dimensions, wind tunnel testing*; **equipment:** *colour wheel, decibel meter, weigh scale*).

A5. Reporting and Presenting

By the end of this course, students will:

A5.1 use technical terminology correctly when documenting, reporting on, and presenting design projects (e.g., *vernier caliper, orthographic, fillet, bisect, construction lines, shears*);

A5.2 compare reporting styles and formats (e.g., **styles:** *American Psychological Association [APA], Modern Language Association [MLA]*; **formats:** *portfolio, journal, logbook, technical report, reflection paper*);

A5.3 describe and use the appropriate tools (e.g., *word processor, multimedia hardware, display board, image board [lifestyle, mood, styling, usage]*) and formats (e.g., *oral or multimedia presentation, technical report*) for documenting, reporting, and presenting design ideas and results;

A5.4 demonstrate an understanding of the components of a technical report (e.g., *design brief, criteria and constraints, idea development, planning, design analysis, evaluation, technical drawings, design summary*).

B. TECHNOLOGICAL DESIGN SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use appropriate strategies and tools to research and manage design projects and related activities;
- B2.** apply appropriate methods for generating and graphically representing design ideas and solutions;
- B3.** create and test models and/or prototypes, using a variety of techniques, tools, and materials;
- B4.** use a variety of formats and tools to create and present reports summarizing the design process and to reflect on decisions made during the process.

SPECIFIC EXPECTATIONS

B1. Researching and Managing Projects

By the end of this course, students will:

- B1.1** use various research methods and strategies to gather, organize, and interpret design information from appropriate resources (*e.g., building codes, Machinery's Handbook, interviews, union contracts*);
- B1.2** use project management strategies and tools (*e.g., project evaluation and review technique [PERT], time sheets, critical path analysis*) to plan and organize finances, human resources, and materials for projects and related activities.

B2. Developing and Representing Design Ideas

By the end of this course, students will:

- B2.1** use freehand sketches to help brainstorm initial design concepts for a project;
- B2.2** apply mathematical and scientific concepts and skills as required in the course of designing projects;
- B2.3** differentiate between artistic and technical criteria for a design;
- B2.4** produce hand-drafted and/or computer-based working drawings and other technical drawings of design solutions, using industry-recognized drafting standards and conventions;
- B2.5** apply principles and elements of graphic design to enhance design ideas in concept drawings and/or presentation-quality drawings.

B3. Making and Testing Models and Prototypes

By the end of this course, students will:

- B3.1** create design models (*e.g., conceptual, physical, virtual, theoretical*) and/or functional prototypes (*e.g., proof-of-principle, functional*) using appropriate tools (*e.g., fabric shears, computer-aided design [CAD] software, computer numerical control [CNC] machine tools*), equipment (*e.g., computer, cut-out pattern*), procedures (*e.g., tracing, forming, assembly*), and materials (*e.g., matt board, starch, linen, neoprene*);
- B3.2** use appropriate metric and imperial measuring tools, scales (*e.g., metric: 1:10, 1 cm:1 m or 1:100, 1:500; imperial: 1/2":1' or 1:24*), and proportion techniques when creating and assessing models and/or prototypes;
- B3.3** assess products and/or processes on the basis of student-developed criteria (*e.g., adherence to specifications, effectiveness, cost, durability, appearance, estimated labour and material costs, cultural sensitivities, effect on the environment*).

B4. Reporting and Presenting

By the end of this course, students will:

- B4.1** create and present reports summarizing design choices and the steps taken in the design process, using a variety of formats and tools (*e.g., word processor, presentation software, interactive white board, web pages*);
- B4.2** report and reflect on decisions made throughout the design process, using a variety of oral and/or written formats.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of environmentally responsible design practices, and apply them in the technological design process and related activities;
- C2.** describe the relationship between society and technological development.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding of environmental issues that affect the design of products and/or processes (*e.g., gasoline consumption, pollution, greenhouse gases, habitat loss, extinction of species, depletion of natural resources*);
- C1.2** describe, advocate, and apply best practices for conserving energy and other resources when designing a product or process (*e.g., reuse or recycle lumber and other materials; use materials with recycled content; use wood glue instead of hot glue; use renewable energy sources, high-efficiency motors and appliances, and passive heating and cooling of buildings*);
- C1.3** describe ways to reduce the waste produced by the manufacture and use of products (*e.g., cutting patterns that minimize leftover materials, use of materials that are easily recycled, energy-management controls in electronic equipment*), and apply such practices when developing and building prototypes.

C2. Technology and Society

By the end of this course, students will:

- C2.1** research and compare technological eras (*e.g., agricultural, industrial, information*), and describe ways in which societal needs influenced these eras;
- C2.2** research and describe cases where technological design has improved the quality of living (*e.g., fireproofing, prosthetic limbs, air purifiers, catalytic converters*);
- C2.3** demonstrate an understanding of ways in which history, trends, culture, and geography have inspired technological design.

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** describe and apply health, safety, and environmental practices related to technological design;
- D2.** identify career opportunities in fields related to technological design, and describe the training and education required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe the importance of health and safety laws, regulations, and standards that apply to technological design (*e.g., regulations and standards from the Occupational Health and Safety Act, Canadian Standards Association [CSA], Ontario Building Code, and Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** adhere to appropriate personal and environmental health and safety standards and procedures with respect to processes, materials, tools, equipment, and facilities throughout the design process and when performing related activities (*e.g., use protective equipment; set tool and equipment guards properly; ensure adequate ventilation and ergonomic seating and other workplace arrangements; follow safe operating procedures; keep work areas clean and organized; store materials and dispose of wastes properly*);
- D1.3** use protective clothing, gear, and equipment appropriately (*e.g., dust mask, safety glasses*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify a variety of career opportunities related to technological design (*e.g., architect, architectural technologist, drafts person, interior designer*);
- D2.2** describe the educational and training pathways (i.e., selection of secondary and post-secondary courses, programs, and learning experiences) and entry requirements (*e.g., portfolio, internship*) for careers related to technological design;
- D2.3** research and report on professional associations and unions for technical designers (*e.g., Professional Engineers of Ontario [PEO], Ontario Association of Chartered Industrial Designers of Ontario [OACID], Ontario Association of Certified Engineering Technicians and Technologists [OACETT], Ontario Association of Architects [OAA], Association of Registered Interior Designers of Ontario [ARIDO]*), and describe how these organizations affect jobs in technological design (*e.g., working conditions, job security*);
- D2.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the technological design industry (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D2.5** demonstrate an understanding of and apply the Essential Skills that are important for success in the technological design industry, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, measurement and calculation*);
- D2.6** demonstrate an understanding of and apply the work habits that are important for success in the technological design industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, initiative, customer service, entrepreneurship*);
- D2.7** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in technological design (*e.g., work logs, skills checklist, sketches, drawings, photographs of models and prototypes*), and explain why having a current portfolio is important for career development and advancement.